

# INSURANCE MANAGEMENT SYSTEM (IMS)

## DDL Queries:

```
mysql> create database insurancedb;
Query OK, 1 row affected (0.05 sec)

mysql> use insurancedb;
Database changed
mysql> create table address_details(address_id int primary key,h_no varchar(6),city varchar(50),addressline1 varchar(50),state
-> varchar(50),pin varchar(50));
Query OK, 0 rows affected (0.06 sec)

mysql> show tables;
+-----+
| Tables_in_insurancedb |
+-----+
| address_details       |
+-----+
1 row in set (0.03 sec)

mysql> select * from insurancedb;
ERROR 1146 (42S02): Table 'insurancedb.insurancedb' doesn't exist
mysql> select * from address_details;
Empty set (0.02 sec)

mysql> create table user_details(user_id int primary key,firstname varchar(50),lastname varchar(50),email varchar(50),mobilen
-> varchar(50),address_id int references address_details(address_id),dob date);
Query OK, 0 rows affected (0.03 sec)

mysql> create table ref_policy_types(policy_type_code varchar(10) primary key,policy_type_name varchar(50));
Query OK, 0 rows affected (0.03 sec)

mysql> create table policy_sub_types(policy_type_id varchar(10) primary key,policy_type_code varchar(10) references
-> ref_policy_types(policy_type_code),description varchar(50),yearsofpayements int,amount double,maturityperiod
-> int,maturityamount double,validity int);
Query OK, 0 rows affected (0.03 sec)

mysql> create table user_policies(policy_no varchar(20) primary key,user_id int references user_details(user_id),date_registered
-> date,policy_type_id varchar(10) references policy_sub_types(policy_type_id) );
Query OK, 0 rows affected (0.02 sec)

mysql> create table policy_payments(receipno int primary key,user_id int references user_details(user_id),policy_no varchar(20)
-> references user_policies(policy_no),dateofpayment date,amount double,fine double);
Query OK, 0 rows affected (0.03 sec)
```

# DML Queries:

```
mysql> insert into address_details values(1,'6-21','hyderabad','kphb','andhrapradesh',1254);
Query OK, 1 row affected (0.02 sec)

mysql> insert into address_details values(2,'7-81','chennai','seruseri','tamilnadu',16354);
Query OK, 1 row affected (0.01 sec)

mysql> insert into address_details values(3,'3-71','lucknow','street','uttarpradesh',86451);
Query OK, 1 row affected (0.00 sec)

mysql> insert into address_details values(4,'4-81','mumbai','iroli','maharashtra',51246);
Query OK, 1 row affected (0.01 sec)

mysql> insert into address_details values(5,'5-81','bangalore','mgroad','karnataka',125465);
Query OK, 1 row affected (0.01 sec)

mysql> insert into address_details values(6,'6-81','ahamadabad','street2','gujarat',125423);
Query OK, 1 row affected (0.01 sec)

mysql> insert into address_details values(7,'9-21','chennai','sholinganur','tamilnadu',654286);
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_details values(1111,'raju','reddy','raju@gmail.com','9854261456',4,'1986-4-11');
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_details values(2222,'vamsi','krishna','vamsi@gmail.com','9854261463',1,'1990-4-11');
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_details values(3333,'naveen','reddy','naveen@gmail.com','9854261496',4,'1985-3-14');
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_details values(4444,'raghava','rao','raghava@gmail.com','9854261412',4,'1985-9-21');
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_details values(5555,'harsha','vardhan','harsha@gmail.com','9854261445',4,'1992-10-11');
Query OK, 1 row affected (0.01 sec)

mysql> insert into ref_policy_types values('58934','car');
Query OK, 1 row affected (0.01 sec)

mysql> insert into ref_policy_types values('58539','home');
Query OK, 1 row affected (0.01 sec)

mysql> insert into ref_policy_types values('58683','life');
Query OK, 1 row affected (0.00 sec)
```

```
mysql> insert into policy_sub_types values('6893','58934','theft',1,5000,null,200000,1);
Query OK, 1 row affected (0.01 sec)

mysql> insert into policy_sub_types values('6894','58934','accident',1,20000,null,200000,3);
Query OK, 1 row affected (0.01 sec)

mysql> insert into policy_sub_types values('6895','58539','fire',1,50000,null,500000,3);
Query OK, 1 row affected (0.00 sec)

mysql> insert into policy_sub_types values('6896','58683','anandhlife',7,50000,15,1500000,null);
Query OK, 1 row affected (0.00 sec)

mysql> insert into policy_sub_types values('6897','58683','sukhlife',10,5000,13,300000,null);
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_policies values('689314',1111,'1994-4-18','6896');
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_policies values('689316',1111,'2012-5-18','6895');
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_policies values('689317',1111,'2012-6-20','6894');
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_policies values('689318',2222,'2012-6-21','6894');
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_policies values('689320',3333,'2012-6-18','6894');
Query OK, 1 row affected (0.01 sec)

mysql> insert into user_policies values('689420',4444,'2012-4-09','6896');
Query OK, 1 row affected (0.03 sec)

mysql> insert into policy_payments values(121,4444,'689420','2012-4-09',50000,null);
Query OK, 1 row affected (0.01 sec)

mysql> insert into policy_payments values(345,4444,'689420','2013-4-09',50000,null);
Query OK, 1 row affected (0.01 sec)

mysql> insert into policy_payments values(225,1111,'689316','2012-5-18',20000,null);
Query OK, 1 row affected (0.01 sec)

mysql> insert into policy_payments values(227,1111,'689314','1994-4-18',50000,null);
Query OK, 1 row affected (0.00 sec)

mysql> insert into policy_payments values(300,1111,'689317','2012-6-20',20000,null);
```

```
mysql> insert into policy_payments values(100,1111,'689314','1995-4-10',50000,null);
Query OK, 1 row affected (0.01 sec)

mysql> insert into policy_payments values(128,1111,'689314','1996-4-11',50000,null);
Query OK, 1 row affected (0.00 sec)

mysql> insert into policy_payments values(96,1111,'689314','1997-4-18',50000,200);
Query OK, 1 row affected (0.01 sec)

mysql> insert into policy_payments values(101,1111,'689314','1998-4-09',50000,null);
Query OK, 1 row affected (0.00 sec)

mysql> insert into policy_payments values(105,1111,'689314','1999-4-08',50000,null);
Query OK, 1 row affected (0.01 sec)

mysql> insert into policy_payments values(120,1111,'689314','2000-4-05',50000,null);
Query OK, 1 row affected (0.00 sec)

mysql> insert into policy_payments values(367,2222,'689318','2012-6-21',20000,null);
Query OK, 1 row affected (0.01 sec)

mysql> insert into policy_payments values(298,3333,'689320','2012-6-18',20000,null);
Query OK, 1 row affected (0.01 sec)
```

## Solve the following Queries:

1. Write a query to display the policytypeid,policytypename,description of all the car's policy details.

```
mysql> SELECT ps.policy_type_id,rpt.policy_type_name,ps.description FROM policy_sub_types ps JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code WHERE rpt.policy_type_name = 'car';
```

| policy_type_id | policy_type_name | description |
|----------------|------------------|-------------|
| 6893           | car              | theft       |
| 6894           | car              | accident    |

2 rows in set (0.00 sec)

2. Write a query to display the policytypecode,no of policies in each code with alias name NO\_OF\_POLICIES.

```
mysql> SELECT policy_type_code,COUNT(*) AS NO_OF_POLICIES FROM policy_sub_types GROUP BY policy_type_code;
```

| policy_type_code | NO_OF_POLICIES |
|------------------|----------------|
| 58934            | 2              |
| 58539            | 1              |
| 58683            | 2              |

3 rows in set (0.01 sec)

3. Write a query to display the userid,firstname,lastname, email,mobilenos who are residing in Chennai.

```
mysql> SELECT u.user_id, u.firstname, u.lastname, u.email, u.mobilenos FROM user_details u JOIN address_details a ON u.address_id = a.address_id WHERE a.city = 'chennai';
```

Empty set (0.00 sec)

4. Write a query to display the userid, firstname lastname with alias name USER\_NAME,email,mobilenos who has taken the car policies.

```
mysql> SELECT DISTINCT u.user_id,CONCAT(u.firstname,' ',u.lastname) AS USER_NAME,u.email,u.mobilenos FROM user_details u JOIN user_policies up ON u.user_id = up.user_id JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code WHERE rpt.policy_type_name = 'car';
```

| user_id | USER_NAME     | email            | mobilenos  |
|---------|---------------|------------------|------------|
| 1111    | raju reddy    | raju@gmail.com   | 9854261456 |
| 2222    | vamsi krishna | vamsi@gmail.com  | 9854261463 |
| 3333    | naveen reddy  | naveen@gmail.com | 9854261496 |

3 rows in set (0.01 sec)

5. Write a query to display the userid, firstname, last name who has taken the car policies but not home policies

```
mysql> SELECT DISTINCT u.user_id, u.firstname, u.lastname FROM user_details u WHERE u.user_id IN (SELECT up.user_id FROM user_policies up JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code WHERE rpt.policy_type_name = 'car') AND u.user_id NOT IN (SELECT up.user_id FROM user_policies up JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code WHERE rpt.policy_type_name = 'home');
```

| user_id | firstname | lastname |
|---------|-----------|----------|
| 2222    | vamsi     | krishna  |
| 3333    | naveen    | reddy    |

2 rows in set (0.01 sec)

6. Write a query to display the policytypecode, policytype name which policytype has maximum no of policies.

```
mysql> SELECT rpt.policy_type_code,rpt.policy_type_name FROM ref_policy_types rpt JOIN policy_sub_types ps ON rpt.policy_type_code = ps.policy_type_code JOIN user_policies up ON ps.policy_type_id = up.policy_type_id GROUP BY rpt.policy_type_code, rpt.policy_type_name ORDER BY COUNT(*) DESC LIMIT 1;
```

| policy_type_code | policy_type_name |
|------------------|------------------|
| 58934            | car              |

1 row in set (0.00 sec)

7. Write a query to display the userid, firstsname, lastname, city state whose city is ending with 'bad'.

```
mysql> SELECT u.user_id, u.firstname, u.lastname, a.city, a.state FROM user_details u JOIN address_details a ON u.address_id = a.address_id WHERE a.city LIKE '%bad';
```

| user_id | firstname | lastname | city      | state         |
|---------|-----------|----------|-----------|---------------|
| 2222    | vamsi     | krishna  | hyderabad | andhrapradesh |

1 row in set (0.00 sec)

8. Write a query to display the userid, firstname, lastname, policyno, dateregistered who has registered before may 2012.

```
mysql> SELECT u.user_id, u.firstname, u.lastname, up.policy_no, up.date_registered FROM user_details u JOIN user_policies up ON u.user_id = up.user_id WHERE up.date_registered < '2012-05-01';
```

| user_id | firstname | lastname | policy_no | date_registered |
|---------|-----------|----------|-----------|-----------------|
| 1111    | raju      | reddy    | 689314    | 1994-04-18      |
| 4444    | raghava   | rao      | 689420    | 2012-04-09      |

2 rows in set (0.00 sec)

9. Write a query to display the userid, firstname, lastname who has taken more than one policies.

```
mysql> SELECT u.user_id, u.firstname, u.lastname FROM user_details u JOIN user_policies up ON u.user_id = up.user_id GROUP BY u.user_id, u.firstname, u.lastname HAVING COUNT(up.policy_no) > 1;
```

| user_id | firstname | lastname |
|---------|-----------|----------|
| 1111    | raju      | reddy    |

1 row in set (0.00 sec)

10. Write a query to display the policytypecode, policytypename, policytypeid, userid, policyno whose maturity will fall in the month of august 2013.

```
mysql> SELECT rpt.policy_type_code, rpt.policy_type_name, ps.policy_type_id, up.user_id, up.policy_no FROM user_policies up JOIN policy_sub_types ps ON u.p.policy_type_id = ps.policy_type_id JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code WHERE DATE_ADD(up.date_registered, INTERVAL ps.maturityperiod YEAR) BETWEEN '2013-08-01' AND '2013-08-31';
```

Empty set (0.01 sec)

11. Write a query to display the policytypecode, policytypename, policytypeid whose maturity amount is the double than the total paid amount.

```
mysql> SELECT rpt.policy_type_code, rpt.policy_type_name, ps.policy_type_id FROM policy_sub_types ps JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code JOIN user_policies up ON ps.policy_type_id = up.policy_type_id JOIN policy_payments pp ON up.policy_no = pp.policy_no GROUP BY rpt.policy_type_code, rpt.policy_type_name, ps.policy_type_id, ps.maturityamount HAVING ps.maturityamount = 2 * SUM(pp.amount);
```

Empty set (0.01 sec)

12. Write a query to display the userid, total amount paid by the customer with alias name total\_amount.

```
mysql> SELECT user_id, SUM(amount) AS total_amount FROM policy_payments GROUP BY user_id;
```

| user_id | total_amount |
|---------|--------------|
| 1111    | 390000       |
| 4444    | 100000       |
| 3333    | 20000        |
| 2222    | 20000        |

4 rows in set (0.00 sec)

13. Write a query to display the user\_id, policy\_no, total amount paid by the customer for the each policies.

```
mysql> SELECT user_id,policy_no,SUM(amount) AS total_amount FROM policy_payments GROUP BY user_id, policy_no;
```

| user_id | policy_no | total_amount |
|---------|-----------|--------------|
| 1111    | 689314    | 350000       |
| 4444    | 689420    | 100000       |
| 1111    | 689316    | 20000        |
| 3333    | 689320    | 20000        |
| 1111    | 689317    | 20000        |
| 2222    | 689318    | 20000        |

6 rows in set (0.00 sec)

14. Write a query to display the user\_id, policy\_no, balance\_amount for each policies.

```
mysql> SELECT up.user_id,up.policy_no,ps.maturityamount - SUM(pp.amount) AS balance_amount FROM user_policies up JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id JOIN policy_payments pp ON up.policy_no = pp.policy_no GROUP BY up.user_id, up.policy_no, ps.maturityamount;
```

| user_id | policy_no | balance_amount |
|---------|-----------|----------------|
| 1111    | 689314    | 1150000        |
| 4444    | 689420    | 1400000        |
| 1111    | 689316    | 480000         |
| 3333    | 689320    | 180000         |
| 1111    | 689317    | 180000         |
| 2222    | 689318    | 180000         |

6 rows in set (0.01 sec)

15. write a query to display the user\_id,policy\_no, balancepayment years with alias name BALANCE\_YEARS for all the customer for each policies.

```
mysql> SELECT up.user_id,up.policy_no,ps.yearsofpayments - COUNT(pp.receivepno) AS BALANCE_YEARS FROM user_policies up JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id LEFT JOIN policy_payments pp ON up.policy_no = pp.policy_no GROUP BY up.user_id, up.policy_no, ps.yearsofpayments;
```

| user_id | policy_no | BALANCE_YEARS |
|---------|-----------|---------------|
| 1111    | 689314    | 0             |
| 1111    | 689316    | 0             |
| 1111    | 689317    | 0             |
| 2222    | 689318    | 0             |
| 3333    | 689320    | 0             |
| 4444    | 689420    | 5             |

6 rows in set (0.00 sec)

16. Write a query to display the user details userid,firstname,last who has taken car, home and life loans.

```
mysql> SELECT u.user_id, u.firstname, u.lastname FROM user_details u JOIN user_policies up ON u.user_id = up.user_id JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code GROUP BY u.user_id, u.firstname, u.lastname HAVING COUNT(DISTINCT rpt.policy_type_name) = 3;
```

| user_id | firstname | lastname |
|---------|-----------|----------|
| 1111    | raju      | reddy    |

1 row in set (0.00 sec)

17. Write a query to select policy\_type\_code,total amount paid by all the customers with alias name total\_amount for each policy department.

```
mysql> SELECT rpt.policy_type_code,SUM(pp.amount) AS total_amount FROM policy_payments pp JOIN user_policies up ON pp.policy_no = up.policy_no JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code GROUP BY rpt.policy_type_code;
```

| policy_type_code | total_amount |
|------------------|--------------|
| 58683            | 450000       |
| 58539            | 20000        |
| 58934            | 60000        |

3 rows in set (0.00 sec)

18. Write a query to select user\_id,user\_name,policy\_type\_code,policy\_type\_id of users who has registered more than one policy type unde same policy code.

```
mysql> SELECT u.user_id,CONCAT(u.firstname,' ',u.lastname) AS user_name,rpt.policy_type_code,ps.policy_type_id FROM user_details u JOIN user_policies up ON
u.user_id = up.user_id JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_
code WHERE (u.user_id, rpt.policy_type_code) IN(SELECT up.user_id, ps.policy_type_code FROM user_policies up JOIN policy_sub_types ps ON up.policy_type_id =
ps.policy_type_id GROUP BY up.user_id, ps.policy_type_code HAVING COUNT(DISTINCT ps.policy_type_id) > 1)ORDER BY u.user_id, rpt.policy_type_code;
Empty set (0.00 sec)
```

19. Write a query to display the policy\_type\_code,policytype name in which policy department has min number of policies registered.

```
mysql> SELECT rpt.policy_type_code,rpt.policy_type_name FROM ref_policy_types rpt JOIN policy_sub_types ps ON rpt.policy_type_code = ps.policy_type_code JOI
N user_policies up ON ps.policy_type_id = up.policy_type_id GROUP BY rpt.policy_type_code, rpt.policy_type_name ORDER BY COUNT(*) ASC LIMIT 1;
+-----+-----+
| policy_type_code | policy_type_name |
+-----+-----+
| 58539            | home             |
+-----+-----+
1 row in set (0.00 sec)
```

20. Write a query to display the user\_id,user\_name, address,phoneno,policytypecode,policytypeid,policytypename, who has complemented all payments for the policies.

Mysql

```
mysql> SELECT u.user_id,CONCAT(u.firstname,' ',u.lastname) AS user_name, CONCAT(a.h_no,' ', a.addressline1, ' ', a.city, ' ', a.state) AS address,u
.mobilenno AS phoneno,rpt.policy_type_code, ps.policy_type_id,rpt.policy_type_name FROM user_details u JOIN address_details a ON u.address_id=a.address_id
JOIN user_policies up ON u.user_id = up.user_id JOIN policy_sub_types ps ON up.policy_type_id = ps.policy_type_id JOIN ref_policy_types rpt ON ps.policy_type_
code = rpt.policy_type_code JOIN policy_payments pp ON up.policy_no = pp.policy_no WHERE ps.yearsofpayements IS NOT NULL GROUP BY u.user_id,user_name,addr
ess,u.mobilenno, rpt.policy_type_code,ps.policy_type_id,rpt.policy_type_name,ps.yearsofpayements HAVING COUNT(pp.receivepno) >= ps.yearsofpayements;
+-----+-----+-----+-----+-----+-----+-----+
| user_id | user_name | address | phoneno | policy_type_code | policy_type_id | policy_type_name |
+-----+-----+-----+-----+-----+-----+-----+
| 1111    | raju reddy | 4-81, iroli, mumbai, maharashtra | 9854261456 | 58683 | 6896 | life |
| 1111    | raju reddy | 4-81, iroli, mumbai, maharashtra | 9854261456 | 58539 | 6895 | home |
| 3333    | naveen reddy | 4-81, iroli, mumbai, maharashtra | 9854261496 | 58934 | 6894 | car |
| 1111    | raju reddy | 4-81, iroli, mumbai, maharashtra | 9854261456 | 58934 | 6894 | car |
| 2222    | vamsi krishna | 6-21, kphb, hyderabad, andhrapradesh | 9854261463 | 58934 | 6894 | car |
+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

21. write a query to display the user\_id, user\_name, address,phoneno,policytypecode,policytypeid,policytypename,date ofd register who has registered latest 2.

```
mysql> SELECT u.user_id,CONCAT(u.firstname,' ',u.lastname) AS user_name,a.addressline1, u.mobilenno,rpt.policy_type_code,ps.policy_type_id,rpt.policy_t
ype_name,up.date_registered FROM user_policies up JOIN user_details u ON up.user_id = u.user_id JOIN address_details a ON u.address_id = a.address_id JOIN p
olicy_sub_types ps ON up.policy_type_id = ps.policy_type_id JOIN ref_policy_types rpt ON ps.policy_type_code = rpt.policy_type_code ORDER BY up.date_registe
red DESC LIMIT 2;
+-----+-----+-----+-----+-----+-----+-----+
| user_id | user_name | addressline1 | mobilenno | policy_type_code | policy_type_id | policy_type_name | date_registered |
+-----+-----+-----+-----+-----+-----+-----+
| 2222    | vamsi krishna | kphb | 9854261463 | 58934 | 6894 | car | 2012-06-21 |
| 1111    | raju reddy | iroli | 9854261456 | 58934 | 6894 | car | 2012-06-20 |
+-----+-----+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```